

Unified Military Analytics and Comparison Dashboard – DV

Official Technical Documentation – Modules 5 & 6

Project File Name: global_military_firepower_2025.pbix

Tool Used: Microsoft Power BI Desktop

MODULE 5 – Full Dashboard Development

5.1 Quick Stats Dashboard

Objective:

To provide a high-level summary of global military power distribution using key KPIs and regional filters.

Key Components Implemented:

- Total Countries (Distinct count of country)
- Average Power Index (Average of overall_power_index)
- Top Country (Rank 1 based on overall_rank)
- Total Defense Budget (Sum of defense_budget_usd)
- Total Military Manpower (Sum of total_military_manpower)

Visualizations Used:

- Clustered Bar Chart: Overall Power Index by Country (Top 10)
- Pie Chart: Average Assets per Capita by Region
- Table: Country, Rank, Defense Budget

Filters Implemented:

- Continent (Dropdown slicer)
- Region (Dropdown slicer)
- NATO Membership (Binary slicer)

Outcome:

The Quick Stats dashboard provides an executive-level overview of global military standings with interactive filtering capability.

5.2 Nation Overview Dashboard

Objective:

To provide detailed military profile analysis for a selected country.

Core KPIs:

- Overall Rank
- Total Military Manpower
- Defense Budget (USD)
- Total Military Assets
- Assets per Capita

Domain-Level Analysis Implemented:

- Manpower Composition (Donut Chart)
- Land Power Composition (Tanks, Armored Vehicles)
- Air Power Composition (Fighter, Transport, Attack Aircraft)
- Naval Fleet Composition (Carriers, Destroyers, Frigates)
- Economy vs Defense Spending Comparison
- Domain Contribution to Military Power (Normalized Indices)

Outcome:

Nation Overview delivers a comprehensive single-country performance breakdown across manpower, land, air, naval, and financial dimensions.

MODULE 6 – Compare Powers & Coalition Builder

6.1 Compare Powers Dashboard

Objective:

To enable side-by-side comparison between two selected countries across key military indicators.

Comparison Metrics Implemented:

- Overall Rank Comparison

- Defense Budget Comparison
- Military Manpower Comparison
- Military Power Efficiency (Power Index per Budget)
- Budget per Soldier
- Air / Land / Naval Dependency Ratios
- Total Military Aircraft, Tanks, Armored Vehicles Comparison

Visual Techniques Used:

- Clustered Column Charts
- Horizontal Bar Charts
- Stacked Bar Charts for Domain Dependency
- Derived KPI Measures using DAX

Outcome:

The Compare Powers dashboard allows dynamic dual-country evaluation using synchronized slicers and consistent aggregation logic.

6.2 Coalition Builder Dashboard

Objective:

To analyze combined military strength of multiple selected countries as a coalition.

Coalition KPIs Implemented:

- Coalition Total Manpower (SUM aggregation)
- Coalition Total Assets
- Coalition Defense Budget
- Coalition Air Support Ratio
- Coalition Naval Projection Ratio
- Coalition Average Budget per Country

Visualizations Implemented:

- Donut Chart: Coalition Asset Distribution

- Donut Chart: Coalition Domain Contribution (%)
- Stacked Bar Chart: Aircraft Composition within Coalition
- Bar Chart: Manpower by Coalition Member

Dynamic Behavior:

All coalition metrics update automatically based on multi-country slicer selection.

6.3 Dashboard Integration & Navigation

- Page Navigator used for seamless transitions.
- Slicers synced across pages where required.
- Consistent formatting, layout alignment, and KPI standards maintained.
- Interactive filtering validated across all dashboards.

Final Outcome:

The global_military_firepower_2025.pbix file contains four fully integrated dashboards: Quick Stats, Nation Overview, Compare Powers, and Coalition Builder. The solution satisfies all Module 5 and Module 6 requirements including comparison logic, coalition aggregation, dynamic filtering, and seamless navigation.